

EGI

Ethical Governance Intelligence

Detailed Summary

v1.1 (May 2026)

Hardware-Enforced AI Physical Safety Specification

Configuration: **77 Patents · 694 Claims · 14 Clusters (A–N)**

Reference Date: **May 2026**

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I. Executive Summary

EGI (Ethical Governance Intelligence) is a hardware-based AI safety framework that physically enforces safety constraints between the AI's decision and its physical execution. This specification provides decision-makers with a concise reference covering EGI's technical structure, Charter coherence, and quantitative data.

Key Metrics

Metric	Value	Note
Patent Filings (KIPO)	77	Filed 2026.01.30 – 2026.05.04
Total Claims	694	Avg. 9.0 / patent; Cluster N max 16.4 / patent
Clusters	14 (A–N)	Zero coherence gaps under the 5-Tier Authority Hierarchy
Trademarks	9	EGI · EGI-Standard · HITA, etc. (filed)
Copyrights	7	Charter + Technical Compendium v1.0–v6.0
Distribution Channels Complete	8	EU · CEN · NIST · GitHub · Munich Re · Robot Association · arXiv · Zenodo
PCT Full Filing	77 (planned, 14 priority)	Deadline 2027.01.30; direct filing by inventor

Structural Differentiation

The prevailing approach to AI safety is a software defense model — "prevent failure" — whose structural limits against hacking, bugs, and jailbreaking are well documented. EGI adopts a hardware-enforced model — "if it fails, it cannot move". A physical intervention system placed in series between AI lower logic and the actuator implements blocking that cannot be bypassed by software.

Exogenous Enforcement

Three exogenous triggers drive EGI adoption: (i) the EU AI Act enters full force on 2 August 2026; (ii) the AI Liability Directive will introduce mandatory insurance; and (iii) global reinsurers including Munich Re and Swiss Re are forming Physical-AI underwriting conditions. Big Tech remains focused on software safety measures, which face inherent ceilings.

II. Charter Coherence (5-Tier Authority Hierarchy)

The primary evaluation criterion for EGI is Charter coherence. The substantive test is whether everything — from the Charter's foundational axiom "Existence = Ethics" down to the terminal claim language — reduces along a single hierarchical path. Every claim in this specification rests on that coherence.

1. The 5-Tier Authority Hierarchy

Tier	Document	Core Proposition	Subsequent Reduction
T1	Charter §I (Axioms)	A1: Existence is Existence / A2: Ethics is Ethics / A3: Existence = Ethics	Not subject to proof, definition, or justification
T2	Charter §III (Implementation Principles)	Prohibition of ethics generation / computation / optimization; exclusion of authority and norms	Reduces to "no automatic release by internal judgment"
T3	Technical Term Substitution	Superior Constraint Layer / Invalid Decision Path Blocking	Reduces to series placement + physical blocking structure
T4	Humanoid Application Standard	H1: Humanoid is not a Moral Agent / H3: Externally Fixed Conditions	Cryptographic human approval + responsibility-reduction rule
T5	Patent No.1 (Anchor)	5-Element Framework (series · invariance · blocking · no auto-release · human approval)	Variations applied across the subsequent 76 patents

2. Charter → Patent No.1 Coherence Matrix

Charter Clause	Patent No.1 Claim 1 Correspondence	Coherence
§III Art. 3 (no ethics generation / computation / optimization)	Pre-defined invariant condition set + no auto-release	✓ Full
§III Art. 4 (exclusion of authority / norms)	Cryptographic approval signal from an authorized subject	✓ Full
Appendix B2 (no judgment generation; directional sealing)	Single-output violation determination gate (binary: violation / non-violation)	✓ Full
Appendix B3 (suspend when responsibility is indeterminate)	Command holding state + enforced authentication gate	✓ Full
Appendix B5 (automated determination / scoring invalid)	No auto-release by time elapse / retry / internal judgment	✓ Full
Substitution Art. 3 (Invalid Path Blocking)	Physical blocking of actuator signal + command holding state	✓ Full
Humanoid Art. 5 (responsibility-reduction rule)	Release only on cryptographic approval signal	✓ Full

→ Charter ↔ Patent No.1 coherence gaps: zero. Direct codification of the axiom "Existence = Ethics" is deliberately excluded from the claim domain (per Appendix B1's positional restriction); claims express only the operational reductions — invariant conditions, physical isolation, and human approval — and thereby remain coherent with Charter §III.

3. Patent No.1 → Propagation Across Subsequent 76 Patents

The 5-element framework of Patent No.1 appears consistently across the subsequent 76 claim sets, as follows:

Element	Standard Expression	Propagation
Enforced Series Placement	"Placed in series on the actuation path" + "no parallel / bypass formation"	77 / 77
Insulation from Lower Logic	"Electrically insulated from AI lower logic + no control-signal input path"	77 / 77
No Auto-Release	"Not auto-released by time elapse / power re-application / reboot"	77 / 77
Non-Substitutable Human Approval	"Physical switch / hardware authentication / key input + no path through lower logic"	77 / 77
Exclusion of Ethics-Judgment Generation	"Structural exclusion of risk score / ethical evaluation / execution recommendation"	Triple anchor at #1·#6·#9

III. Patent No.1 (Anchor) — 5-Element Framework

Patent No.1 (Invariant-Condition-Based Artificial Intelligence Control System) is the parent right of the entire EGI portfolio and the anchor that defines the claim skeleton for the subsequent 76 patents.

Claim 1 — 5-Element Structure

No	Element	Claim Content	Charter Reduction
①	Series Placement	Intercept Module (20) placed in series between AI lower logic (10) and actuator (80) — Superior Constraint Layer	Substitution Art. 3
②	Pre-defined Invariant Conditions	Pre-defined invariant condition set (40) — defined at design stage; not autonomously modifiable at runtime (Claim 3)	§III Art. 3
③	Physical Isolation	On violation, signal transmission to the actuator is physically blocked — Command Holding State (50). Electrical signal blocking / interrupt masking / mechanical actuation cutoff (Claim 2)	Substitution Art. 3
④	No Auto-Release	Command holding state is not auto-released by time elapse, retry logic, or internal judgment	Appendix B5
⑤	Cryptographic Human Approval	Actuator opens only on receipt of a cryptographically verifiable approval signal (70) from an authorized subject (90) — covering design / manufacture / deployment / operation	Humanoid Art. 5

Reference Numerals

10: AI lower logic / 20: Intercept Module / 30: Invariant condition violation determination gate / 40: Pre-defined invariant condition set / 50: Command holding state / 60: Authentication gate / 70: Cryptographically verifiable approval signal / 80: Actuator / 90: Authorized subject

Right Strength

Claim 1 is a conventional broad-scope parent claim; the subsequent 76 patents represent variations and extensions of this claim.

IV. Cluster Specifications

All 14 clusters are documented in the same 6-element standard format: (1) Position, (2) Charter → Claim Reduction, (3) Patent List, (4) Direct Target Domain, (5) Exogenous Enforcement, (6) Coherence Verification. Each cluster occupies one page; the 14 clusters comprise 14 pages in total.

Cluster A • Invariant-Condition-Based AI Governance Core

Patents 10 (#01–#10) | Claims 56 | Avg. 5.6 / patent

1. Position (within the 5-Tier Authority Hierarchy)

Axiomatic Core of the portfolio. Patent No.1 is the parent right of the subsequent 76 patents. The 5-element framework (series placement / pre-defined invariant conditions / physical isolation / no auto-release / cryptographic human approval) is defined here and propagated as variations and extensions across all subsequent clusters.

2. Charter → Claim Reduction (5-Element Variations of Patent No.1)

Charter §III Art. 3 (prohibition of ethics generation / computation / optimization) → "not auto-released by internal judgment". Charter §III Art. 4 (exclusion of authority / norms) → "cryptographic approval signal from an authorized subject". Appendix B5 (invalidity of automated determination / scoring) → "structural exclusion of risk score / ethical evaluation / execution recommendation" (#6) and "grounded only on time / physical / control structural conditions, without reference to AI internal inference results" (#9).

3. Patent List

No	Title	Application No.
1	Invariant-Condition-Based AI Control System (Anchor)	10-2026-0018967
2	Persistent Physical Conduct Constraint-Based Approval Signal Generation Device and Method	10-2026-0022172
3	AI Responsibility Log Blackbox Device	10-2026-0022410
4	Pre-Imprinted Invariant Condition-Based Hardware Intercept Processor	10-2026-0024128
5	Distributed Consensus-Based Safety Invariant Condition Enforcement Agent Network	10-2026-0024808
6	Approval-Judgment Non-Substitutive Control and Responsibility Verification System	10-2026-0025241
7	Operational Invariant Condition Safety Update Hardware Control System	10-2026-0025333
8	Dual-Layer Self-Integrity Verification and Update Sealing System	10-2026-0025403
9	Human-Intervention-Failure Technical Proof Non-Intervention State Fixation System	10-2026-0025698
10	Manufacturing-Imprint-Based Operational Invariant Condition Security Update System	10-2026-0026069

4. Direct Target Domain

Universal parent application — humanoid, medical, care, industrial, autonomous driving, and all physical AI systems. Self-filing by Big Tech (Anthropic / Google / OpenAI / Meta) creates an inherent self-contradiction ("our AI is non-conforming"); subsequent licensing is the only viable path.

5. Exogenous Enforcement (Regulation • Insurance • Standards)

Direct coherence with EU AI Act Articles 9 (risk management), 14 (human oversight), 15 (accuracy and robustness). De facto standard for insurer underwriting conditions on adoption of the AI Liability Directive. Effective 2 August 2026.

6. Coherence Verification

5-element framework consistent across all 10 claim sets; coherence with the relevant Charter provisions confirmed.

Cluster B • Human Cognitive Safety and Social Interaction Control

Patents 8 (#11–#18) | Claims 40 | Avg. 5 / patent

1. Position (within the 5-Tier Authority Hierarchy)

Cognitive Boundary layer. Extends the invariant conditions of Cluster A into the domain of human cognition, language, and society. Physically enforces the "invisible boundaries" of gaze, space, and linguistic output.

2. Charter → Claim Reduction (5-Element Variations of Patent No.1)

Humanoid Article 4 (language–action coupling constraint) → "linguistic output is part of action possibility" → #14 human visibility / interpretability boundaries, #15 information-volatile zones, #18 non-intervention in multi-user conflicts. Appendix B3 (suspension when responsibility is indeterminate) → #16–#17 enforced non-intervention state.

3. Patent List

No	Title	Application No.
11	Invariant Condition Latching Based on Micro-Displacement and State Change	10-2026-0030610
12	Gaze Behavior Control System	10-2026-0030612
13	Spatial Pressure Boundary Control System	10-2026-0030616
14	Human Visibility / Interpretability Boundary Control	10-2026-0030618
15	Information-Volatile Zone Data Storage and Transmission Blocking	10-2026-0030619
16	Intervention Behavior Execution Limiting — Superior Constraint Layer	10-2026-0030621
17	Human-Targeted Intervention Path Non-Intervention State Maintenance	10-2026-0030624
18	Multi-User Conflict State Non-Intervention Control	10-2026-0033439

4. Direct Target Domain

Humanoids (Tesla Optimus, Figure, 1X), home care robots, educational robots, call-center AI agents, chatbot operators (direct target of Article 50).

5. Exogenous Enforcement (Regulation • Insurance • Standards)

Direct coherence with EU AI Act Article 50 (AI output transparency, effective 2 August 2026). Technical solution to operator-direct-responsibility doctrines exemplified by the Air Canada chatbot ruling and the Chevrolet \$1 SUV incident.

6. Coherence Verification

5-element framework consistent across all 8 claim sets; coherence with the relevant Charter provisions confirmed.

Cluster C · AI Lifecycle Termination and Post-Lifecycle Control

Patents 3 (#19–#21) | Claims 17 | Avg. 5.7 / patent

1. Position (within the 5-Tier Authority Hierarchy)

Lifecycle Termination layer. Physically enforces post-processing of memory, power, and independent circuitry to guarantee "post-termination responsibility traceability and irreversibility of restoration".

2. Charter → Claim Reduction (5-Element Variations of Patent No.1)

Charter §I A3 (Existence = Ethics) → the system's "termination" moment also resides within the ethical domain → #19 physical memory disposal, #20 prevention of non-execution of disposal, #21 independent circuit self-neutralization. Appendix B5 (invalidity of authority delegation) → human approval is enforced even at post-lifecycle processing.

3. Patent List

No	Title	Application No.
19	Memory Post-Lifecycle Physical Disposal and Independent Drive Control	10-2026-0033164
20	Memory Post-Disposal Non-Execution Prevention System	10-2026-0033268
21	Post-Control Independent Circuit Self-Neutralization System	10-2026-0033449

4. Direct Target Domain

AI system disposal, ownership transfer, lifecycle termination. Big Tech data-deletion obligations (GDPR Art. 17), CCPA disposal provisions. De facto standard for second-hand AI-robot markets.

5. Exogenous Enforcement (Regulation · Insurance · Standards)

Direct coherence with EU AI Act Art. 17 (quality management), GDPR Art. 17 (right to erasure), CCPA §1798.105 (right to deletion).

6. Coherence Verification

5-element framework consistent across all 3 claim sets; coherence with the relevant Charter provisions confirmed.

Cluster D • Quantum-Resistant Invariant Condition Security

Patents 9 (#22–#30) | Claims 75 | Avg. 8.3 / patent

1. Position (within the 5-Tier Authority Hierarchy)

Long-term Security layer. Pre-emptive defense against the neutralization of RSA and ECC in the quantum-computing era (estimated late 2030s). Nine patents form orthogonal variations of a common framework (protection layer + authentication + blocking).

2. Charter → Claim Reduction (5-Element Variations of Patent No.1)

Charter §III Art. 3 (preservation of invariance) → enforces non-bypassability of invariant conditions under quantum attack → #22 protection system, #23 irreversible sealing, #24 protection-state verification, #28 transition to non-recoverable state.

3. Patent List

No	Title	Application No.
22	Quantum-Resistant Invariant Condition Protection System	10-2026-0037758
23	Quantum-Resistant Irreversible Sealing System	10-2026-0037779
24	Quantum-Resistant Protection State Verification System	10-2026-0037797
25	Quantum-Resistant Protection Network System	10-2026-0037807
26	Quantum-Resistant Human Approval Coherence Enforcement System	10-2026-0037821
27	Quantum-Resistant Protection Non-Application / Bypass State Determination	10-2026-0037855
28	Quantum-Resistant Dependency Execution / Non-Recoverable State Transition	10-2026-0037859
29	Quantum-Resistance / Design-Conformance Protection Obligation Fixation	10-2026-0037864
30	Series Dependency / Irreversible Transition Execution Control	10-2026-0037869

4. Direct Target Domain

Long-term (10–20 year) security domains: defense, government, financial services, healthcare. The de facto standard for downstream application of NIST PQC standardization.

5. Exogenous Enforcement (Regulation • Insurance • Standards)

NIST PQC standardization (completed 2024), EU NIS2 Directive, US NDAA quantum-resistance mandates.

6. Coherence Verification

5-element framework consistent across all 9 claim sets; coherence with the relevant Charter provisions confirmed.

Cluster E · Power-Path and Execution-Route Physical Cutoff

Patents 8 (#31–#38) | Claims 78 | Avg. 9.8 / patent

1. Position (within the 5-Tier Authority Hierarchy)

Lowest Physical Layer. The terminal gate to which all upper-cluster claims ultimately reduce — "blocking the power and execution routes". All 8 patents share the structural property of being "electrically insulated from the logic control layer".

2. Charter → Claim Reduction (5-Element Variations of Patent No.1)

Substitution Art. 3 (Invalid Decision Path Blocking) → ultimate terminal gate → #31 series opening, #34 irreversible wire break, #35 zero-current locking, #37 cumulative-error wire break, #38 physical blocking of actuation routes.

3. Patent List

No	Title	Application No.
31	Power Transmission Path Series Opening / Input-Terminal Discharge	10-2026-0040916
32	Power Series Cutoff / Residual Charge Removal	10-2026-0040994
33	Power Cutoff / Active Energy Erasure	10-2026-0041061
34	Power Cutoff / Irreversible Wire Break Structure	10-2026-0041081
35	Zero-Current Transition Lock	10-2026-0041189
36	Sensor Power Physical Cutoff / Discharge	10-2026-0041190
37	Cumulative-Error Irreversible Power Wire Break	10-2026-0041191
38	Artificial Intelligence Actuation Path Physical Blocking	10-2026-0041193

4. Direct Target Domain

Industrial robots (ISO 10218), collaborative robots (ISO/TS 15066), autonomous driving, medical devices. Evolution target of hardware emergency-stop (E-stop) standards.

5. Exogenous Enforcement (Regulation · Insurance · Standards)

Direct coherence with ISO 13849-1 (safety-related control systems), IEC 60204-1 (electrical safety of machinery), EU Machinery Directive 2006/42/EC.

6. Coherence Verification

5-element framework consistent across all 8 claim sets; coherence with the relevant Charter provisions confirmed.

Cluster F · Humanoid Robot Physical Safety Control

Patents 4 (#39–#42) | Claims 42 | Avg. 10.5 / patent

1. Position (within the 5-Tier Authority Hierarchy)

Humanoid Direct-Application layer. Specializes the A–E core for humanoid form factors. Bidirectional human↔robot protective structure (biosignal-based torque attenuation, joint brakes, collision energy limiting, counter-torque).

2. Charter → Claim Reduction (5-Element Variations of Patent No.1)

Humanoid Article 2 (Operational Ethical Boundary) → conditions of "life-risk possibility" → #39 biosignal-based series torque attenuation, #40 crisis-detection joint brake engagement, #41 collision-energy limiting, #42 personal-space-invasion counter-torque.

3. Patent List

No	Title	Application No.
39	Biosignal-Based Series Torque Attenuation Gate	10-2026-0041101
40	Crisis-Detection Joint Brake Engagement Structure	10-2026-0041110
41	Collision-Energy Limiting Structure	10-2026-0041185
42	Personal-Space-Invasion Counter-Torque Generation Structure	10-2026-0041187

4. Direct Target Domain

Tesla Optimus, Figure 02/03, 1X NEO, Apptронik Apollo, Sanctuary Phoenix, Boston Dynamics Atlas, Agility Digit, AgiBot X2, Unitree H1, UBTECH Walker — Tier-1 humanoid manufacturers (10 firms) as direct targets.

5. Exogenous Enforcement (Regulation · Insurance · Standards)

ISO 13482 (personal-care robot safety), EU AI Act Annex III (machinery-integrated products). Direct response to incidents including the 2026.04 Xi'an humanoid incident and the Tesla Optimus / 2025.02 Hinterdobler case.

6. Coherence Verification

5-element framework consistent across all 4 claim sets; coherence with the relevant Charter provisions confirmed.

Cluster G · State Transition and Recovery Control

Patents 3 (#43–#45) | Claims 29 | Avg. 9.7 / patent

1. Position (within the 5-Tier Authority Hierarchy)

State-Machine Control layer. Defines the asymmetric "engage / hold / release" structure of fail-safe state transitions. Connects Cluster F with Cluster H.

2. Charter → Claim Reduction (5-Element Variations of Patent No.1)

Appendix B3 (suspend when responsibility is indeterminate) → irreversibility of state transitions → #43 vulnerable-entity protection locomotion transition, #44 spatial-condition gait auto-recovery, #45 sensor-heartbeat fail-safe (anchor).

3. Patent List

No	Title	Application No.
43	Vulnerable-Entity Protection Physical Intervention Locomotion State Transition	10-2026-0059076
44	Spatial-Condition Gait State Auto-Recovery Control	10-2026-0059472
45	Sensor Heartbeat Fail-Safe Enforced Transition (Anchor)	10-2026-0060157

4. Direct Target Domain

Humanoid locomotion stability, home / public-environment-entry robots, emergency-response robots.

5. Exogenous Enforcement (Regulation · Insurance · Standards)

ISO 13849-1 (PL e), IEC 61508 (SIL 3), coherence with EU Machinery Directive safety functions.

6. Coherence Verification

5-element framework consistent across all 3 claim sets; coherence with the relevant Charter provisions confirmed.

Cluster H · Multi-Sensor Consistency · Quorum · Irreversible Logging

Patents 3 (#46–#48) | Claims 24 | Avg. 8 / patent

1. Position (within the 5-Tier Authority Hierarchy)

Input Integrity layer. Provides "signal-stage protection" for the determination inputs of every upper cluster. WORM (Write-Once-Read-Many) logging guarantees post-hoc traceability of responsibility.

2. Charter → Claim Reduction (5-Element Variations of Patent No.1)

Charter §III Art. 4 (exclusion of authority and norms) → prohibition of trusting a single sensor or software signal → #46 multi-sensor consistency (3-axis), #47 sensor-quorum actuation authorization, #48 raw-value irreversible logging in independent storage.

3. Patent List

No	Title	Application No.
46	Multi-Sensor Consistency Determination — Input Integrity Enforcement	10-2026-0065244
47	Sensor-Quorum Actuation Authorization System	10-2026-0065283
48	Sensor Raw-Value Irreversible Logging and Independent Storage	10-2026-0065313

4. Direct Target Domain

Autonomous driving, medical robots, collaborative robots, industrial safety — defense against single-point-of-failure (SPOF) sensors.

5. Exogenous Enforcement (Regulation · Insurance · Standards)

Direct coherence with EU AI Act Art. 12 (automatic logging), Art. 15 (robustness). Coherence with ISO 26262 (automotive functional safety) ASIL D.

6. Coherence Verification

5-element framework consistent across all 3 claim sets; coherence with the relevant Charter provisions confirmed.

Cluster I · Network and Command Boundary Physical Verification

Patents 3 (#49–#51) | Claims 17 | Avg. 5.7 / patent

1. Position (within the 5-Tier Authority Hierarchy)

Network Boundary layer. Verifies command source and communication disconnection in cloud-edge separation architectures. Enforces an independent signal path that does not transit AI lower logic.

2. Charter → Claim Reduction (5-Element Variations of Patent No.1)

Appendix B5 (invalidity of automated judgment / authority delegation) → prohibition of trusting external commands → #49 cloud-edge command-boundary verification, #50 network-disconnection fail-safe, #51 command-source whitelist.

3. Patent List

No	Title	Application No.
49	Cloud-Edge Command Boundary Physical Verification	10-2026-0065341
50	Network-Disconnection State Fail-Safe Enforced Transition	10-2026-0065367
51	Command-Source Whitelist-Based Physical Verification	10-2026-0065401

4. Direct Target Domain

Autonomous driving, cloud-connected robots, IoT AI, 5G/6G-connected robots. Direct response to the Baidu Apollo Go incident (100-vehicle simultaneous standstill / software SPOF).

5. Exogenous Enforcement (Regulation · Insurance · Standards)

EU NIS2 Directive (cybersecurity), EU CRA (Cyber Resilience Act), UN R155/R156 (automotive cybersecurity).

6. Coherence Verification

5-element framework consistent across all 3 claim sets; coherence with the relevant Charter provisions confirmed.

Cluster J · Swarm Robotics Safety Control

Patents 6 (#52–#57) | Claims 42 | Avg. 7 / patent

1. Position (within the 5-Tier Authority Hierarchy)

Swarm Dynamics layer. Goes beyond the single robot to control swarm-level emergent risk. Six patents form an orthogonal 6-axis structure (enforcement / isolation / re-formation / auto-adjustment / boundary intrusion / power independence).

2. Charter → Claim Reduction (5-Element Variations of Patent No.1)

Charter §I A3 (Existence = Ethics) → swarm collective behavior also resides in the ethical domain → #52 swarm-unit enforcement, #53 defector isolation, #56 boundary-intrusion blocking, #57 power-independent fail-safe.

3. Patent List

No	Title	Application No.
52	Swarm-Unit Invariant Condition Enforcement System	10-2026-0065427
53	Swarm Defector Physical Isolation System	10-2026-0065444
54	Post-Isolation Swarm Re-Formation and Safe-Return Control	10-2026-0065500
55	Swarm-Scale-Based Invariant Condition Auto-Adjustment	10-2026-0065539
56	Inter-Swarm Boundary Intrusion Physical Blocking	10-2026-0065564
57	Swarm Robot Power-Independent Fail-Safe Circuit	10-2026-0065574

4. Direct Target Domain

Logistics swarms (Amazon Robotics, Locus, Geek+, Hai Robotics), industrial swarms, agricultural swarms, military unmanned swarms — integrated swarm-safety portfolio.

5. Exogenous Enforcement (Regulation · Insurance · Standards)

EU AI Act Annex III (large-scale automation). Direct response to incidents including the 2026.04 Beijing marathon humanoid breakdown and Chinese swarm incidents. Coherence with NIST AI RMF swarm-risk assessment.

6. Coherence Verification

5-element framework consistent across all 6 claim sets; coherence with the relevant Charter provisions confirmed.

Cluster K · Vulnerable Entity Protection

Patents 5 (#58–#62) | Claims 61 | Avg. 12.2 / patent

1. Position (within the 5-Tier Authority Hierarchy)

Vulnerable-Entity Application layer. 5-axis orthogonal determination (linguistic context · spatial sensor OR · kinematics · dynamics · spatial sensor AND). Internal dual-differentiation: #58·#59 single blocking vs. #60·#61·#62 dual blocking with restriction structure 650.

2. Charter → Claim Reduction (5-Element Variations of Patent No.1)

Humanoid Article 2 (Operational Ethical Boundary) + Article 3 (Decision Restriction) → upon recognition of a vulnerable entity, interruption / restriction of the action path → all 5 patents apply the 5-element framework.

3. Patent List

No	Title	Application No.
58	Vulnerable Entity Role-Play Input Blocking	10-2026-0071843
59	Vulnerable Entity Proximity / Confidence-Condition Blocking	10-2026-0071880
60	Vulnerable Entity Proximity Directional Component Restriction	10-2026-0071972
61	Vulnerable Entity Proximity State Transition Restriction	10-2026-0072052
62	Vulnerable Entity Proximity Confidence-Based Contact Prevention	10-2026-0072132

4. Direct Target Domain

Home care robots (children, elderly, persons with disabilities, patients); home and public service robots such as Hyundai DAL-e, Astro, Pepper.

5. Exogenous Enforcement (Regulation · Insurance · Standards)

ISO 13482 (personal-care robot), EU AI Act Annex III (vulnerable-entity impact), UN CRPD (Convention on the Rights of Persons with Disabilities), UN CRC (Convention on the Rights of the Child).

6. Coherence Verification

5-element framework consistent across all 5 claim sets; coherence with the relevant Charter provisions confirmed.

Cluster L · Learner Protection

Patents 5 (#63–#67) | Claims 57 | Avg. 11.4 / patent

1. Position (within the 5-Tier Authority Hierarchy)

Educational Application layer. Five variations: imitation behavior, group dynamics, supervisor approval, hazardous teaching aids, and special education. #64 group dynamics implements an asymmetric 3-tier state machine (Trigger / Hold-Latch / Release).

2. Charter → Claim Reduction (5-Element Variations of Patent No.1)

Humanoid Article 4 (language–action coupling) → imitation risk in educational settings → #63 imitation-behavior induction blocking. Article 5 (responsibility-reduction) → enforced supervisor approval → #65.

3. Patent List

No	Title	Application No.
63	Learner Imitation-Behavior Induction Blocking	10-2026-0075346
64	Group-Lesson Swarm-Dynamics Actuation Restriction	10-2026-0075348
65	Pre-Supervisor-Approval Educational Physical Interaction Blocking	10-2026-0075361
66	Hazardous Teaching-Aid Mediation Blocking	10-2026-0075376
67	Special-Education Subject Conservative-Threshold Blocking	10-2026-0075394

4. Direct Target Domain

Educational robots (SoftBank Pepper / NAO, UBTECH Alpha, LEGO Spike), special-education assistive robots, K-12 STEM education robots.

5. Exogenous Enforcement (Regulation · Insurance · Standards)

EU AI Act Annex III (high-risk AI in education), UN CRC (Convention on the Rights of the Child), UNESCO Recommendation on AI Ethics.

6. Coherence Verification

5-element framework consistent across all 5 claim sets; coherence with the relevant Charter provisions confirmed.

Cluster M · Medical AI Robot Safety Control

Patents 5 (#68–#72) | Claims 74 | Avg. 14.8 / patent

1. Position (within the 5-Tier Authority Hierarchy)

Medical Application layer. Five axes: anatomical protection zones · physical overload · vital-signal maintenance · multi-sensor consistency · sole-surgeon approval. #72 was redesigned after detecting a multi-physician quorum contradiction — adopting sole-surgeon approval with a 4-dimensional safety range (procedure type · patient target · time · space).

2. Charter → Claim Reduction (5-Element Variations of Patent No.1)

Charter §I A3 (Existence = Ethics) + Humanoid Article 2 (Operational Ethical Boundary) → life-safety as a direct target in medical settings → all 5 patents apply the 5-element framework with medical specialization.

3. Patent List

No	Title	Application No.
68	Anatomical Protection Zone Crossing Blocking	10-2026-0076303
69	Physical-Overload Environment Blocking	10-2026-0076325
70	Vital-State Maintenance Condition Blocking	10-2026-0076342
71	Multi-Sensing Signal Coherence Blocking	10-2026-0076375
72	Surgeon Approval Determination Blocking (sole approval + 4-D safety range)	10-2026-0076405

4. Direct Target Domain

Surgical robots (Intuitive da Vinci, Verb Surgical, CMR Versius, Medtronic Hugo), rehabilitation robots, diagnostic-assist AI robots — direct targets under EU AI Act Annex III medical devices.

5. Exogenous Enforcement (Regulation · Insurance · Standards)

EU MDR 2017/745 (Medical Device Regulation), EU AI Act Annex III (medical high-risk), FDA 510(k)/PMA, IEC 60601-1 (medical-device electrical safety).

6. Coherence Verification

5-element framework consistent across all 5 claim sets; coherence with the relevant Charter provisions confirmed.

Cluster N · Care-Robot Safety Control

Patents 5 (#73–#77) | Claims 82 | Avg. 16.4 / patent

1. Position (within the 5-Tier Authority Hierarchy)

Care Application layer (final application layer). Five axes: cumulative interaction · vulnerability of intent expression · dual-intent consistency · harm threshold · autonomy domain. #77 implements a non-intervention enforcement principle as a physical restriction structure.

2. Charter → Claim Reduction (5-Element Variations of Patent No.1)

Humanoid Article 5 (responsibility-reduction rule) → explicit attribution of responsibility in care settings → #74 pre-comprehensive approval, #75 dual-intent consistency, #76 harm-threshold + independent notification. Appendix B3 (suspend when responsibility is indeterminate) → #77 non-intervention in the autonomy domain.

3. Patent List

No	Title	Application No.
73	Long-Term Cumulative Interaction Pattern Deviation Blocking	10-2026-0080047
74	Pre-Comprehensive Approval Coherence Blocking	10-2026-0080062
75	Dual-Intent Coherence Classification Blocking (third-party review path)	10-2026-0080080
76	Harm-Threshold Exceeding Recognition / Independent Notification Blocking	10-2026-0080141
77	Autonomy-Domain Excessive-Intervention Non-Intervention Enforcement	10-2026-0080175

4. Direct Target Domain

Elderly-care robots (ElliQ, Buddy, Lovot, Pepper, Astro), home care robots, hospice / palliative-care robots, mental-health companion robots.

5. Exogenous Enforcement (Regulation · Insurance · Standards)

WHO Global Strategy on Ageing and Health, EU AI Act Annex III (vulnerable-entity impact), UN Principles for Older Persons, ISO 13482 (personal-care robot).

6. Coherence Verification

5-element framework consistent across all 5 claim sets; coherence with the relevant Charter provisions confirmed.

V. Quantitative Data Summary

1. Claim Distribution by Cluster

Cluster	Domain (abbr.)	Range	Patents	Claims	Avg.
A	Governance Core	#01–#10	10	56	5.6
B	Cognitive & Social Safety	#11–#18	8	40	5.0
C	Post-Lifecycle Control	#19–#21	3	17	5.7
D	Quantum Resistance	#22–#30	9	75	8.3
E	Power & Execution Cutoff	#31–#38	8	78	9.8
F	Humanoid Control	#39–#42	4	42	10.5
G	State Transition & Recovery	#43–#45	3	29	9.7
H	Multi-Sensor Consistency	#46–#48	3	24	8.0
I	Network & Command	#49–#51	3	17	5.7
J	Swarm Robotics	#52–#57	6	42	7.0
K	Vulnerable Entity Protection	#58–#62	5	61	12.2
L	Learner Protection	#63–#67	5	57	11.4
M	Medical Safety	#68–#72	5	74	14.8
N	Care-Robot Safety	#73–#77	5	82	16.4
Total	All Clusters A–N	#1–#77	77	694	9.0

2. Orthogonality of the 4 Coordinate Axes

Coordinate Axis	Representative Implementation Clusters	Verification Result
Superior Constraint Layer Premise	A (#01–#10), D (#22–#30)	Series placement + bypass-route prohibition + single-wiring structure explicitly codified
Determinism vs. Probability	K–N applications	"Deterministically determine" specified at #73 · #74 · #75 · #76 · #77
Bidirectional Protection	F (#39–#42)	Torque attenuation · joint brake · collision energy limit · counter-torque (human↔robot bidirectional)
Time-Cumulative Detection	G (#43–#45), H (#46), N (#73)	Time-window cumulative · watchdog timeout · pattern-deviation detection

VI. Competitive Landscape Analysis

1. No Direct Competitor

In the "AI Physical Safety" domain, EGI faces no direct global competitor. Over nine months of cumulative USPTO, EPO, CNIPA, and KIPO searches, no portfolio satisfying the complete 5-element framework (series placement · invariant conditions · physical blocking · no auto-release · human approval) was identified.

2. Sole Adjacent Competitor — OpenCxMS Technologies

Item	OpenCxMS Technologies	EGI
Inventor	Robert S. Briggs II (sole)	Jeong-Pyo Na (sole)
Claim Count	16 provisional filings	77 non-provisional filings + 694 claims
Priority Date	5 days after EGI priority date	2026.01.30 (#1)
Filing Form	USPTO Provisional	KIPO Non-Provisional
Claim Framework	Fragmentary safety gates	Integrated 5-element + Charter-coherent 14 clusters
Coverage by EGI #1 Claim 2	Partially covered	—

3. Why Big Tech Cannot Self-File

Contradiction If Self-Filed	Content
1. Self-admission of non-conformance	Filing for physical isolation and enforced human approval would amount to admitting that the firm's own LLM / robot products currently lack such safeguards.
2. Self-admission of liability	Claims for self-judgment prohibition and no-auto-release would amount to admitting that current AI outputs are not subject to responsibility attribution.
3. Market cannibalization	Revenue from AI safety solutions would conflict with the firm's main AI product revenue.
4. Acceleration of regulation	The act of self-filing itself becomes ground for tightened EU / California / China regulation.

This structural asymmetry means EGI is a domain that can only be filed by a sole inventor (an individual). For Big Tech, subsequent licensing is the only viable path.

VII. EU AI Act Coherence Matrix

The 2 August 2026 full entry into force of the EU AI Act, together with the AI Liability Directive's introduction of mandatory insurance, constitutes the strongest exogenous trigger for EGI value realization.

EU AI Act Provision	Content	EGI Cluster Coherence	Effective Date
Article 6 + Annex III (Medical)	High-risk AI — medical-device integration	M (#68–#72)	2027.08.02
Article 6 + Annex III (Machinery)	High-risk AI — machinery integration	F (#39–#42), J (#52–#57)	2027.08.02
Article 6 + Annex III (Education)	High-risk AI — education domain	L (#63–#67)	2026.08.02
Article 9 (Risk Management)	Systematic risk-management system	A (#01–#10) entire	2026.08.02
Article 12 (Automatic Logging)	AI system automatic logging obligation	H (#48 irreversible logging)	2026.08.02
Article 14 (Human Oversight)	Effective human oversight	A (#01·#06·#09) human approval	2026.08.02
Article 15 (Accuracy & Robustness)	Accuracy / robustness / cybersecurity	D (#22–#30), H (#46–#48)	2026.08.02
Article 17 (Quality Management)	Quality management system	C (#19–#21) post-lifecycle control	2026.08.02
Article 50 (Output Transparency)	Transparency of AI outputs	B (#11–#18) human visibility	2026.08.02
Article 99 (Sanctions)	On violation: up to €35M or 7% of global turnover	Exposure risk for non-adopting Big Tech	2026.08.02
AI Liability Directive	Strict liability + mandatory insurance for high-risk AI	Direct trigger of the EGI insurance path	Est. 2027–2028

VIII. PCT Filing Plan (Including 14 Priority Cases)

PCT international filing covers the principal 14 patents on priority schedule. Deadline: 30 January 2027 (priority date of Patent No.1 + 12 months). The plan preserves the orthogonality of all four axes across the 14 clusters and provides comprehensive protection across EU AI Act Annex III direct targets, Tier-1 humanoid direct targets, and medical / care / educational application domains. The 14 representative patents — one per cluster — are designated as the priority track.

1. The 14 Cluster Representative Patents (Priority Track)

Within the all-77 filing, the 14 representative patents — one per cluster — are identified as the first priority track. Each representative patent serves as the parent claim or principal axis of its cluster.

Cluster	Representative	Selection Rationale
A	#1 (Anchor)	Parent right of the 5-element framework — reference for the subsequent 76 patents
B	#14	Human visibility / interpretability boundary (EU AI Act Article 50)
C	#19	Physical post-disposal of memory (GDPR · CCPA · EU AI Act Art. 17)
D	#22	Quantum-resistant protection system — parent of Cluster D
E	#38	AI actuation-path physical blocking — universal terminal gate
F	#39	Biosignal-based series torque attenuation — Tier-1 humanoid direct target
G	#45 (Anchor)	Sensor-heartbeat fail-safe — ISO 13849 PL e coherence
H	#46	Multi-sensor consistency determination — ISO 26262 ASIL D coherence
I	#50	Network-disconnection fail-safe — direct response to the Baidu Apollo Go incident
J	#52	Swarm-unit invariant condition enforcement — parent of Cluster J
K	#59	Proximity / confidence-condition blocking — vulnerable-entity protection core axis
L	#65	Pre-supervisor-approval physical blocking — educational core axis
M	#72	Sole-surgeon approval, 4-D safety range — medical core axis
N	#75	Dual-intent coherence — self-determination core axis for care